Emiliano Rodriguez

emiliano.rf.7@outlook.com | LinkedIn | Portfolio

Education

Mechatronics Engineering - Robotics Specialization Sept. 2021 – April 2026
Humber Polytechnic, ON - Bachelor of Engineering Honours GPA: 92.2

Work Experience

Mechanical Engineering Co-op Student, Crown Food Service Equipment Jan. 2025 – May 2025

Industrial kitchen equipment manufacturer, handling all stages of production.

- Created engineering drawings of parts, guiding production in processes such as welding, sheet metal forming.
- Created full 3D models in SolidWorks of assemblies and created mechanical drawings.
- Redesign sheet metal products to simplify welding with a collaborative robot, optimizing the design for the robot's capabilities, reducing welding time by 50%

Electromechanical Robotics Technician – Co-op student, F&P Manufacturing May 2024 – Dec. 2024

Automated plant manufacturing auto part for Honda, Jeep, Cadillac from raw materials.

- Diagnosed and resolved problems related to electrical circuits, weld imperfections (GMAW), sensor failures, robotic manipulators, and production issues, ensuring optimal performance and product quality.
- Performed daily troubleshooting and maintenance on 200+ robotic manipulators (Fanuc, Panasonic, Yaskawa)
- Implemented PLC improvements (SIEMENS, Allen Bradley) to prevent quality issues, reducing defective parts.
- Worked closely with QA teams to identify and implement improvements in manufacturing cells.
- Support deployment of vision systems with object detection algorithms for automated QA.

Research Assistant, Humber Polytechnic Jan. 2024 – Mar. 2025

Research in collaboration with Stack Teck, with the goal of designing an automated system to shape heating elements, aiming to reduce lead time by 70%, saving subcontracting cost of nearly $62,000 per year.

- Research and developed prototypes solutions using SolidWorks and 3D printing, iterating designs.
- Developed test setups to determine material properties, guiding inform design decisions based on data.
- Performed calculations to guide the design choices, ensuring long lifespan and accuracy of the system.
- Delivered final design of full system to costumer, including SolidWorks model, documentation, safety systems.

Engineering Projects & Leadership

Electrical Lead, Humber Autonomous Surface Vehicles (ASV) April 2023 – August 2026

Humber Polytechnic - RoboBoat (USA) & NJORD Challenge (Norway)

- Designed, tested, and integrated custom PCBs, electrical architecture following AWG standards.
- Applied FMEA to mitigate electromagnetic interference (EMI), water ingress, and battery depletion.
- Contributed to machining mill machining custom parts, fiberglass composite fabrication, 3D printing, ROS architecture development, LiDAR and depth-camera point-cloud processing, and Gazebo simulation.
- Managed project timelines, coordinated tasks, and led technical meetings.

Capstone project: Electrically Assisted Accessibility Vehicle

Designed and produced custom front wheel drive accessibility vehicle for customer with special needs and one driver, to fit through a 30 in door, with redundant safety features and 300 kg payload capacity.

- Performed beam-bending calculations to meet requirements of drive train/chassis, supported by FEA
- Machined 40+ custom designed parts on mill to a spec of +/- 0.001 inches
- Designed power train, hydraulic steering and electrical circuits, with corresponding drawings.
- Developed smart safety braking using LiDAR sensor, bringing vehicle to stop in case of an imminent collision.

Other Skills

- Proficiency in programming languages including Python, C, C++, PLC (ladder logic, FBD, etc.).
- Skilled with Microsoft Office products (PowerPoint, Excel, Project, Word).
- Experience with several manufacturing processes (CNC, milling, lathe, welding, sheet metal working).
- Data analytics using Minitab, MATLAB, Python PANDAS.
- Proficiency in English and Spanish.